University of Notre Dame

Department of Computer Science and Engineering

CSE 60745

Graph Learning: Techn & Apps

HW3 — Part (B)

Assignment Date:	
Sumission Date:	
Student Name:	

Q1: Structural Encoding in Graph Transformer

`LapPETransformer` adds Laplacian eigenvector positions to every node once, before the first block; `SPDAttention` adds a learnable per-head scalar, indexed by the pair's hop distance, to every attention score. Two choices in the absolute encoding matter.

Which eigenvectors, and at what scale. The smallest eigenvalue of $L = I - D^{-1/2}AD^{-1/2}$ is 0 with the trivial eigenvector $D^{1/2}\mathbf{1}$, so `precompute` drops it and returns columns $1, \ldots,$ `LAP_K`; a graph with fewer non-trivial eigenvectors is zero-padded (ZINC's smallest molecule has 9 nodes, so exactly 8 exist). Each column is then scaled by $\sqrt{n}$: eigenvector entries are $O(n^{-1/2})$, so unscaled a channel has RMS 0.011 on the 7624-node LastFM graph, where `num_atom_types=0` leaves the encoding as the model's only input.

Sign ambiguity. v and $-v$ are the same eigenvector and the solver picks one arbitrarily, so each graph's channels get an independent random sign flip while training and the identity at evaluation. That is an augmentation *across* graphs, so it is skipped when the batch holds only one — on LastFM it would be noise on the model's only input.

The relative encoding computes all-pairs hop distances by frontier expansion: exact on ZINC, blocked and stopped after `max_dist` hops on LastFM; distances from `max_dist` upwards share one bias anyway, and unreachable pairs keep their own row. The lookup is added to the scores before mask and softmax, $\text{score}_{ij} = q_i^\top k_j/\sqrt{d} + b_{\text{SPD}(i,j)}$, which is Graphormer's form. Distances were checked against a plain BFS, eigenvectors against the dense spectrum of L.

Table 1: ZINC subset, 30 epochs, dense attention; 4 blocks, dim 64, 4 heads, the framework's fixed seed. Epoch selected on validation MAE.

encoding	val MAE	test MAE	best epoch
none (plain `Attention`)	0.6472	0.7013	18
demo: degree centrality + common neighbours	0.4629	0.4957	27
LapPE only (absolute)	0.6205	0.6575	30
SPD only (relative)	0.5373	0.5614	29
LapPE + SPD (cell 14)	**0.5160**	**0.5496**	30

SPD carries almost all of the gain over plain attention (0.7013 to 0.5614 test MAE); LapPE alone reaches 0.6575. Putting LapPE on top of SPD moves it by 0.0119, inside the 0.0200 seed spread of the full-attention row of Table 2, so this budget does not establish a LapPE effect beyond SPD. Both choices were ablated on both datasets, and behave differently. The $\sqrt{n}$ scaling is invisible on ZINC (0.5449 against 0.5496) and worth 0.0508 micro-F1 on LastFM. The sign flip is worth 0.0503 test MAE on ZINC and costs 0.1281 micro-F1 on LastFM, which is why it is conditional.

Implementation (notebook cell 13, with its two helpers)

```
SPD_MAX_DIST = 8          # distances above this share one bias (SPDAttention.max_dist)
DENSE_LIMIT = 1024        # graphs this small are handled with dense linear algebra


def laplacian_eigvecs(edge_index, num_nodes, k=LAP_K):
    """Rows of the k eigenvectors of L = I - D^-1/2 A D^-1/2 with the smallest
    non-trivial eigenvalues, scaled by sqrt(n) so each channel has unit RMS.
    Padded with zero columns when the graph has fewer than k non-trivial eigenvectors."""
    n = num_nodes
    A = sp.coo_matrix((np.ones(edge_index.size(1)), (edge_index[0].numpy(), edge_index[1].numpy())),
                      shape=(n, n)).tocsr()
```

```
    dinv = 1.0 / np.sqrt(np.maximum(np.asarray(A.sum(1)).ravel(), 1.0))
    Ahat = sp.diags(dinv) @ A @ sp.diags(dinv)          # = I - L, same eigenvectors
    m = min(k, n - 1)                                   # non-trivial eigenvectors available
    out = np.zeros((n, k), dtype=np.float32)
    if m > 0:
        if n <= DENSE_LIMIT:
            w, V = np.linalg.eigh(np.eye(n) - Ahat.toarray())           # ascending
        else:                                            # smallest of L = largest of I - L
            v0 = np.random.default_rng(RANDOM_SEED).standard_normal(n)   # fixed ARPACK start
            w, V = sp.linalg.eigsh(Ahat, k=m + 1, which="LA", tol=0, v0=v0)
            w, V = 1.0 - w, V
            order = np.argsort(w)
            w, V = w[order], V[:, order]
        out[:, :m] = V[:, 1:m + 1] * math.sqrt(n)        # skip the trivial eigenvector
    return torch.from_numpy(out)


def hop_distances(edge_index, num_nodes, cap=None):
    """All-pairs hop distance by frontier expansion, as int16 [n, n].

    Unreachable pairs (different connected components) are -1.  With `cap` set the
    expansion stops after `cap` hops and every pair that is further away is reported
    as `cap`; those pairs share the max_dist bias anyway."""
    n = num_nodes
    src, dst = edge_index[0].numpy(), edge_index[1].numpy()
    A = sp.coo_matrix((np.ones(len(src), dtype=np.float32), (src, dst)), shape=(n, n)).tocsr()
    D = np.full((n, n), -1, dtype=np.int16)
    limit = n if cap is None else cap
    if n <= DENSE_LIMIT:
        Ad = np.zeros((n, n), dtype=np.float32)
        Ad[src, dst] = 1.0
        seen = np.eye(n, dtype=bool)
        frontier = seen.copy()
        D[seen] = 0
        for d in range(1, limit + 1):
            frontier = ((frontier.astype(np.float32) @ Ad) > 0) & ~seen
            if not frontier.any():
                break
            D[frontier] = d
            seen |= frontier
    else:
        for s0 in range(0, n, 1024):                     # blocks of sources, columns of F
            s1 = min(s0 + 1024, n)
            F = np.zeros((n, s1 - s0), dtype=np.float32)
            F[np.arange(s0, s1), np.arange(s1 - s0)] = 1.0
            seen = F.astype(bool)
            D[np.arange(s0, s1), np.arange(s0, s1)] = 0
            for d in range(1, limit + 1):
                F = ((A @ F) > 0) & ~seen
                if not F.any():
                    break
                D[s0:s1][F.T] = d
                seen |= F
                F = F.astype(np.float32)
    if cap is not None:                                  # reached later than the cap, not unreachable
        comp = sp.csgraph.connected_components(A, directed=False)[1]
        far = (D < 0) & (comp[:, None] == comp[None, :])
        D[far] = cap
    return torch.from_numpy(D)

class LapPETransformer(GraphTransformer):
    """Absolute encoding: Laplacian eigenvectors, added to the input once."""

    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        # Linear map from the LAP_K eigenvector entries of a node to a node vector of size dim.
        self.lap_proj = nn.Linear(LAP_K, self.dim)

    @staticmethod
    def precompute(edge_index, num_nodes):
        return {"lap": laplacian_eigvecs(edge_index, num_nodes)}

    def inject_absolute(self, h, batch):
```

```
        lap = batch["lap"]
        # Eigenvectors are defined up to sign, so flip each graph's channels at random while
        # training and use the identity at evaluation.  This is an augmentation ACROSS graphs:
        # on a single-graph batch there is nothing to generalise over and the flip is pure
        # input noise, so it is skipped (it costs ~0.12 micro-F1 on LastFM).
        if self.training and batch["num_graphs"] > 1:
            sign = torch.randint(0, 2, (batch["num_graphs"], lap.size(1)),
                                 device=lap.device, dtype=lap.dtype) * 2 - 1
            lap = lap * sign[batch["batch"]]
        return h + self.lap_proj(lap)


class SPDAttention(Attention):
    """Relative encoding: shortest-path distance, one learnable bias per head."""

    def __init__(self, dim, heads, max_dist=SPD_MAX_DIST):
        super().__init__(dim, heads)
        self.max_dist = max_dist
        # one bias per head for each distance 0..max_dist, plus one for unreachable pairs
        self.spd_table = nn.Embedding(max_dist + 2, heads)

    @staticmethod
    def precompute(edge_index, num_nodes):
        cap = None if num_nodes <= DENSE_LIMIT else SPD_MAX_DIST
        return {"spd": hop_distances(edge_index, num_nodes, cap=cap)}

    def bucket(self, d):
        """Hop distance -> row of spd_table: 0..max_dist, and max_dist + 1 if unreachable."""
        d = d.long()
        return torch.where(d < 0, torch.full_like(d, self.max_dist + 1), d.clamp(max=self.max_dist))

    def forward(self, h, enc, mask):
        B, N, D = h.shape
        q, k, v = self.qkv(h).view(B, N, 3, self.heads, D // self.heads).permute(2, 0, 3, 1, 4)
        scores = q @ k.transpose(-1, -2) / math.sqrt(D // self.heads)      # [B, H, N, N]
        bias = self.spd_table(self.bucket(enc["spd"]))                     # [B, N, N, H]
        scores = scores + bias.permute(0, 3, 1, 2)
        scores = scores.masked_fill(~mask[:, None, None, :], float("-inf"))
        weights = torch.softmax(scores, dim=-1).nan_to_num(0.0)
        return self.out((weights @ v).transpose(1, 2).reshape(B, N, D))
```

Q2: Efficient Graph Transformer

`SparseSPDAttention` keeps Q1's parameters and `precompute` and changes only which pairs exist. Five pair sets were compared (Table 2): the graph's edges alone; plus one global virtual node; plus Exphormer's random expander at two widths; and the 2-hop ball — every ordered pair within two hops — plus the virtual node. The submitted class attends to the last of these, picked on validation. Its cost is $\sum_v \deg(v)^2$ rather than n^2: 189 pairs per ZINC molecule, 804,349 on LastFM (1.4% of full). The expander was the a-priori favourite: it spreads global mixing over random edges instead of routing it through one hub, still at $O(n)$ pairs. Both widths finished last on both datasets.

`forward` gathers q, k, v at the endpoints of each pair, adds the inherited SPD bias plus one learnable bias per pair type, normalises with `segment_softmax` over the pairs that share a querying node, and scatters the weighted values back with `index_add_`. The handout points at a `segment_attention` helper that does not exist in PyG 2.8.1; cell 3 imports `softmax as segment_softmax`, and that is what is used. The framework also reports `enc["spd"]` = 0 for pairs touching a virtual node, which would alias them with distance-0 self loops, so those go to the unreachable row of `spd_table` instead. On a complete pair set this path reproduces the dense Q1 attention to 7.5×10^{-8}.

Table 2: Masking strategies on ZINC, same model and budget as Table 1, with LapPE throughout. Pairs are per graph, averaged over the 10,000 training molecules ($n^2 = 557$ on average). The val and test columns are the framework's seed, as cell 18 runs it; the last column is the mean and half-range of the test MAE over that seed and two more, the expander rows redrawing their expander each time. The bolded row has not been reproduced exactly, in either column: three later repeats of that identical configuration — same seed, same weights, same batch order — gave val MAE 0.4318, 0.4418, 0.4418 and test MAE 0.4715, 0.4766, 0.4766, every one of them worse than the draw printed here, because the sparse forward accumulates with `index_add_`, whose order across CPU threads is not fixed. Read any single ZINC number in this table as ±0.01, including the validation column the choice of pair set rests on — that choice is made on the per-seed ranking, which is stable, not on these decimals. All of this compares pair sets inside this framework — 30 epochs, and the bond features `make_batch` carries are never read — rather than reproducing any paper's numbers.

attention pairs	pairs/graph	of full	val MAE	test MAE	3-seed mean
self loops + graph edges	73	13%	0.4918	0.5048	0.5145 ± 0.0133
+ virtual node	120	22%	0.4791	0.5047	0.5092 ± 0.0067
2-hop ball + virtual (submitted)	189	34%	**0.4293**	**0.4677**	0.4909 ± 0.0211
+ 4-regular expander	200	36%	0.5179	0.5551	0.5708 ± 0.0129
+ 8-regular expander	266	48%	0.5213	0.5554	0.5655 ± 0.0119
full attention (Q1)	557	100%	0.5160	0.5496	0.5614 ± 0.0100

The ranking is not the one the pair counts suggest. Restricting attention to the graph beats full attention outright — 0.5145 against 0.5614 test MAE on the three-seed mean — so on 23-atom molecules sparsity is a useful prior, not a sacrifice. Two hops wins again: lowest validation MAE of the five at every seed, and 0.0184 off the runner-up's three-seed test mean, while a comparable random expander loses 0.0799 on it: the pairs have to mean something, and an expander edge means nothing to a molecule. LastFM neither contradicts nor confirms that — its top three finish within 0.008 validation micro-F1, the submitted set second of five — so ZINC decides. Table 3 is what LastFM does settle, the cost: 0.33 s a step against 5.23 s for full attention over all 58,125,376 pairs, a 16× saving, at test micro-F1 0.7827.

Table 3: LastFM (7624 nodes, 27,806 edges, 18 countries), 2 blocks, dim 32, 2 heads, CPU, 8 threads. One training step is the best of 7 forward-backward passes; micro-F1 is `run_node_task` over 200 epochs, selected on validation. The framework's own single-step `benchmark_step` prints 5.24 s and 0.33 s for the two models of cell 19, the same numbers. Full attention is not trained for 200 epochs: at 5.23 s a step it would cost more than every other experiment here together. The framework's peak-memory column prints `n/a on CPU`, so no memory figure is quoted.

attention pairs	pairs attended	of full	one step (s)	test micro-F1
full attention (Q1)	58,125,376	100.000%	5.23	—
self loops + graph edges	63,236	0.109%	0.03	0.7893
+ virtual node	78,485	0.135%	0.04	0.7826
2-hop ball + virtual (submitted)	804,349	1.384%	0.33	**0.7827**
+ 4-regular expander	108,965	0.187%	0.05	0.7678
+ 8-regular expander	139,417	0.240%	0.06	0.7607

Implementation (notebook cell 16)

```
EXPANDER_CYCLES = 2        # union of 2 random Hamiltonian cycles = a 4-regular expander
```

```python
def pair_set(edge_index, num_nodes, n_cycles=EXPANDER_CYCLES, virtual=True, extra=()):
    """The pair set of one graph: self loops, the graph's edges, an expander built as the
    union of n_cycles random Hamiltonian cycles, and one global virtual node (id num_nodes).
    The cycles are drawn from a generator seeded by the graph itself, so they are fixed."""
    n, node = num_nodes, torch.arange(num_nodes)
    gen = torch.Generator().manual_seed((RANDOM_SEED * 1000003 + n * 7919
                                         + int(edge_index.sum())) % (2 ** 31))
    parts = [torch.stack([node, node]), edge_index, *extra]        # self loops + graph edges
    for _ in range(n_cycles if n >= 3 else 0):                     # expander edges
        p = torch.randperm(n, generator=gen)
        q = torch.roll(p, -1)
        parts += [torch.stack([p, q]), torch.stack([q, p])]
    if virtual:                                                    # global node, both directions
        virt = torch.full_like(node, n)
        parts += [torch.stack([node, virt]), torch.stack([virt, node]), torch.tensor([[n], [n]])]
    return coalesce(torch.cat(parts, 1), num_nodes=n + 1)


class SparseSPDAttention(SPDAttention):
    """Your Q1 attention, restricted to a chosen set of node pairs.

    Of the five pair sets measured (see the report) the 2-hop ball plus one global virtual node
    has the lowest validation MAE on ZINC at every seed; on LastFM it is second of five, 0.005
    behind the best.  That is what this class attends to.  pair_set above still builds
    Exphormer's random expander; the expander variants are one argument away and are compared
    in the report."""

    def __init__(self, dim, heads, max_dist=SPD_MAX_DIST):
        super().__init__(dim, heads, max_dist)
        # Exphormer gives each kind of attention edge its own bias: local, expander, global
        self.type_table = nn.Embedding(len(PAIR_TYPES), heads)

    @staticmethod
    def attention_pairs(edge_index, num_nodes):
        """Every ordered pair at most two hops apart, plus one global virtual node."""
        n = num_nodes
        A = sp.coo_matrix((np.ones(edge_index.size(1)), (edge_index[0].numpy(), edge_index[1].numpy())),
                          shape=(n, n)).tocsr()
        ball = ((A + A @ A) > 0).tocoo()                    # pairs at distance 0, 1 or 2
        extra = torch.from_numpy(np.stack([ball.row, ball.col])).long()
        return pair_set(edge_index, num_nodes, n_cycles=0, virtual=True, extra=[extra])

    def forward(self, h, enc, pairs):
        M, D = h.shape
        H, dh = self.heads, D // self.heads
        src, dst = pairs                                        # node dst attends to node src
        q, k, v = self.qkv(h).view(M, 3, H, dh).permute(1, 0, 2, 3)
        score = (q[dst] * k[src]).sum(-1) / math.sqrt(dh)       # [P, H]
        virtual = enc["pair_type"] == PAIR_TYPES.index("virtual")  # their spd arrives as 0,
        d = torch.where(virtual, torch.full_like(enc["spd"], -1), enc["spd"])  # but no distance
        score = score + self.spd_table(self.bucket(d)) + self.type_table(enc["pair_type"])
        alpha = segment_softmax(score, dst, num_nodes=M)        # softmax over each node's pairs
        out = torch.zeros(M, H, dh, device=h.device, dtype=h.dtype)
        out.index_add_(0, dst, alpha.unsqueeze(-1) * v[src])    # [M, H, dh]
        return self.out(out.reshape(M, D))
```

Q3: Hybrid Graph Transformer

Design

`HybridGTLayer` follows GraphGPS (Rampášek et al., NeurIPS 2022; reference [4]): a message-passing branch and a self-attention branch read the *same* input states, and their outputs are added (Figure 1).

Local branch — GINE message passing. Each directed edge (j, i) carries $\mathrm{ReLU}(x_j + e_{ij})$, with e_{ij} a learnable bond-type embedding; a node sums its incoming messages and a two-layer MLP maps $(1+\epsilon)\, x_i + \sum_{j \in \mathcal{N}(i)} \mathrm{ReLU}(x_j + e_{ij})$ to the new state. The aggregation is a sum, as in GIN: mean and max collapse neighbourhood multisets that a sum keeps apart. Bond types enter here and nowhere else, as in GraphGPS, where "the edge features are only passed to the MPNN layer".

Global branch — attention inside one molecule. Ordinary multi-head attention. The states arrive concatenated over the batch, so the branch densifies them with `to_dense_batch` and masks the padding: a node attends only within its own molecule. No structural bias is added inside the attention.

Combination — parallel sum. Each branch gets its own residual connection and LayerNorm, the two streams are added, and a residual two-layer MLP follows: GraphGPS Eqs. (9)–(11) of Appendix D.1, with LayerNorm for BatchNorm so that a node's update cannot depend on which molecules share its batch, and with dropout at zero. The alternative is to stack the branches, letting attention read the MPNN's output. GraphGPS's objection is aimed at the strong form of this, GraphTrans's block of MPNN layers before the transformer, where "these layers could irreparably fail to keep some information in the early stage"; interleaving them inside one layer is the mild form, which the paper does not rule out; I measured it. Summing still gives each branch a gradient path to the input that bypasses the other, so a layer can down-weight either one.

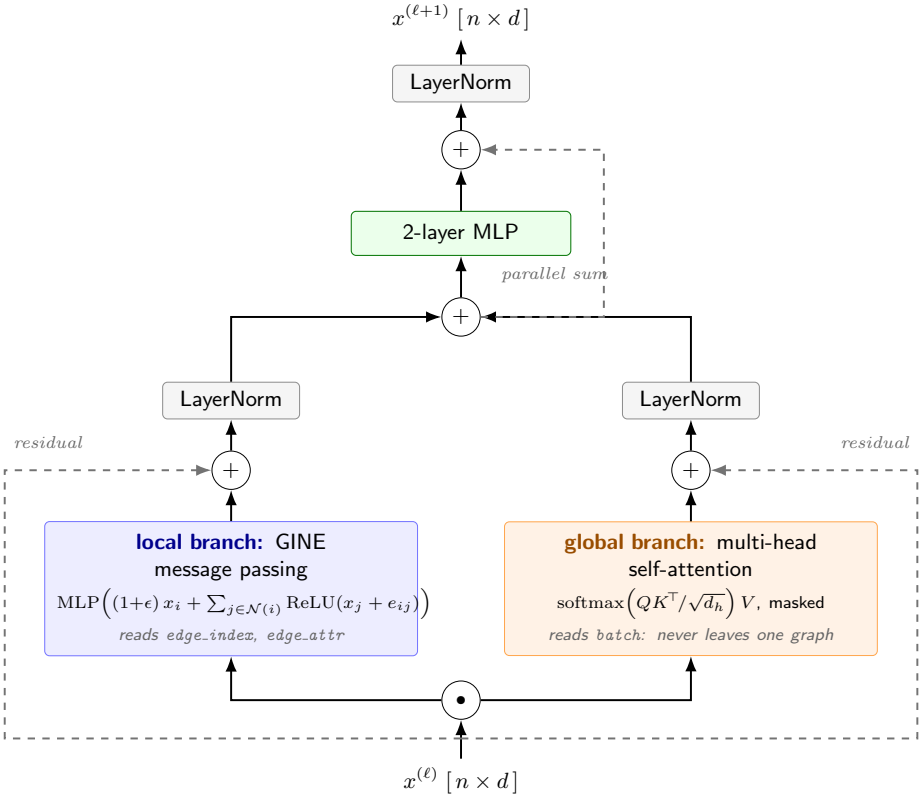


Figure 1: One `HybridGTLayer`. Both branches read $x^{(\ell)}$; dashed arrows are residual connections.

Implementation

`HybridGT` (cell 22) is used unchanged.

```python
class HybridGTLayer(nn.Module):
    """One GraphGPS-style hybrid layer: a local message-passing branch and a
    global self-attention branch, run in parallel on the same input and summed.
    """

    def __init__(self, dim, heads, num_bond_types=4):
        super().__init__()
        self.dim, self.heads = dim, heads
        # local branch: GINE message passing
        self.bond = nn.Embedding(num_bond_types, dim)
        self.eps = nn.Parameter(torch.zeros(1))
        self.mpnn = nn.Sequential(nn.Linear(dim, 2 * dim), nn.ReLU(),
                                  nn.Linear(2 * dim, dim))
        # global branch: multi-head self attention inside each graph
        self.qkv = nn.Linear(dim, 3 * dim)
        self.proj = nn.Linear(dim, dim)
        # fusion
        self.norm_local = nn.LayerNorm(dim)
        self.norm_global = nn.LayerNorm(dim)
        self.ffn = nn.Sequential(nn.Linear(dim, 2 * dim), nn.ReLU(),
                                 nn.Linear(2 * dim, dim))
        self.norm_ffn = nn.LayerNorm(dim)

    def local_branch(self, x, edge_index, edge_attr):
        """GINE: every message carries its bond type, neighbours are summed."""
        src, dst = edge_index
        msg = F.relu(x[src] + self.bond(edge_attr))        # [E, dim]
        agg = torch.zeros_like(x).index_add_(0, dst, msg)  # [num_nodes, dim]
        return self.mpnn((1 + self.eps) * x + agg)

    def global_branch(self, x, batch):
        """Self attention; the padding mask keeps it inside a single graph."""
        h, mask = to_dense_batch(x, batch)          # [B, N, dim], [B, N]
        B, N, D = h.shape
        qkv = self.qkv(h).view(B, N, 3, self.heads, D // self.heads)
        q, k, v = qkv.permute(2, 0, 3, 1, 4)        # each [B, heads, N, D/heads]
        scores = q @ k.transpose(-1, -2) / math.sqrt(D // self.heads)
        scores = scores.masked_fill(~mask[:, None, None, :], float("-inf"))
        weights = torch.softmax(scores, dim=-1).nan_to_num(0.0)
        out = (weights @ v).transpose(1, 2).reshape(B, N, D)
        return self.proj(out)[mask]                 # back to [num_nodes, dim]

    def forward(self, x, edge_index, edge_attr, batch):
        """x          : [num_nodes, dim], node states of the whole batch
        edge_index : [2, E]
        edge_attr  : [E], bond type ids in [0, 4)
        batch      : [num_nodes], graph id of every node
        Returns new node states [num_nodes, dim]."""
        local = self.local_branch(x, edge_index, edge_attr)
        glob = self.global_branch(x, batch)
        h = self.norm_local(x + local) + self.norm_global(x + glob)
        return self.norm_ffn(h + self.ffn(h))
```

Results

Table 4: ZINC, 30 epochs; lower MAE is better, and the epoch is selected on validation MAE. The two baseline rows were measured again for this table rather than copied from Table 1, which reports a single run at the framework's fixed seed; the differences (0.693 against 0.7013, 0.495 against 0.4957) are inside the 0.009 run-to-run noise quoted below. Each row is the mean over independent runs at different seeds, with their standard deviation. Repeating a single configuration unchanged already moves test MAE by a comparable amount on this CPU, so the sd is a run-to-run noise floor rather than a seed effect alone. "params" counts only parameters that receive a gradient, so an ablation is not credited with the branch it never calls. The bold row is the submitted layer, not the best number in the column.

	val MAE		test MAE			
model	mean	sd	mean	sd	params	runs
plain graph transformer (no structural information)	0.649	0.004	0.693	0.002	139.6k	3
demo: degree centrality + common-neighbour bias	0.469	0.015	0.495	0.018	143.9k	3
hybrid GT (cell 22): parallel sum	0.350	0.007	0.364	0.008	211.6k	6
ablation: global branch removed	0.353	0.008	0.367	0.007	144.5k	6
ablation: local branch removed	0.533	0.003	0.573	0.003	143.7k	3
instead: attention reads the MPNN output	0.347	0.006	0.359	0.007	211.6k	6
instead: a learned gate in place of the sum	0.340	0.009	0.355	0.005	244.6k	6

The hybrid cuts the plain transformer's test MAE from 0.693 to 0.364 and beats the demo encoding at 0.495. The ablation says which branch does it: removing attention changes nothing beyond run-to-run noise (0.367), removing message passing costs 0.573. Repeating one configuration unchanged already moves test MAE by 0.009 here, the bar the rest has to clear. Every hybrid run peaks in the final three epochs, so 30 stops it early, and at 100 (one run each) the gap opens the other way — local only 0.251 against 0.283. Stacking the branches matches summing them (0.359), and the gate is the best of the three at 0.355 — by about one noise width, and with 16% more parameters, so I kept the plain sum.

The global branch is not short of range: ZINC molecules have mean diameter 12.5, and four message-passing rounds cover only 51.8% of node pairs. It is short of *identity*. Degree centrality — the absolute encoding `HybridGT` fixes — takes four values and gives half of all atoms the same one, so attention can average over a molecule but hardly tell which atom it reads. Of GraphGPS's three ingredients, the one missing here is the first, the positional/structural encoding.